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Responses to membrane potential-modulating ionic solutions measured by magnetic resonance imaging of cultured cells and *in vivo* rat cortex

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Kyeongseon Min, Sungkwon Chung, Seung-Kyun Lee, Jongho Lee, Phan Tan Toi, Daehong Kim, Jung Seung Lee, Jang-Yeon Park

Department of Electrical and Computer Engineering, Seoul National University, Seoul, Republic of Korea • Department of Physiology, Sungkyunkwan University School of Medicine, Suwon, Republic of Korea • Department of Biomedical Engineering, Sungkyunkwan University, Suwon, Republic of Korea • Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University, Suwon, Republic of Korea • National Cancer Center, Goyang, Republic of Korea

eLife Assessment

The authors show MRI relaxation time changes that are claimed to originate from cell membrane potential changes. This would be a substantial contribution if true because it may provide a mechanism whereby membrane potential changes could be inferred noninvasively. However, the membrane potential manipulations applied here are performed on a slow time scale and are known to induce cell swelling. Cell swelling has been previously shown to affect relaxation time. Experiments could be performed to rule out this hypothesis, but the authors have chosen not to perform these experiments. The study is therefore **useful**, but the evidence is **incomplete**.

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Abstract

Membrane potential plays a crucial role in various cellular functions. However, existing techniques for measuring membrane potential are often invasive or have limited recording depth. In contrast, magnetic resonance imaging (MRI) offers noninvasive imaging with desirable spatial resolution over large areas. This study investigates the feasibility of utilizing MRI to detect responses of cultured cells and *in vivo* rat cortex to membrane potential-modulating ionic solutions by measuring magnetic resonance parameters. Our findings reveal that depolarizing (or hyperpolarizing) ionic solutions increase (or decrease) the T_2 relaxation time, while the ratio of bound to free water proton shows the opposite trend. These findings also suggest a potential approach to noninvasively detect changes in membrane potential using MRI.

Introduction

Membrane potential is a fundamental property of all living cells, influencing crucial cell functions[1] such as neuronal and myocyte excitability, volume control, cell proliferation, and secretion. In the fields of neuroscience, membrane potential is of significant importance, as neural activities arise from the dynamic propagation of this electric potential. From a clinical perspective, deviations from normal membrane potential levels contribute to various diseases, including seizures[2], arrhythmia[3], and hypoglycemia[4]. Given its scientific and clinical importance, effective methods to detect changes in membrane potential have long been sought.

The intracellular recording technique using sharp glass electrodes is a commonly used method to detect changes in membrane potential[5,6]. It provides real-time absolute recordings of membrane potential. Optical imaging is another major approach to detect changes in membrane potential. Voltage-sensitive dyes enable voltage imaging at the cellular level[7]. Fluorescent calcium indicators detect calcium transients associated with neuronal activation[8]. Label-free optical imaging techniques[9] utilize various contrast mechanisms such as cell membrane deformation, which are directly coupled with changes in membrane potential. Despite their efficacy, these methods have limitations when applied to intact biological systems due to their invasive nature, requiring procedures such as craniotomy.

As noninvasive techniques for directly or indirectly detecting brain activation *in vivo*, several imaging modalities have been developed, including electroencephalography (EEG)[10], magnetoencephalography (MEG)[11], and magnetic resonance imaging (MRI)[12]. EEG and MEG detect electric potentials on the scalp and extracranial magnetic fields, respectively, which are directly induced by neuronal activity in the brain. Although these techniques are noninvasive and provide excellent temporal resolution of milliseconds or less, they are constrained by shallow imaging depth and spatial localization challenges[13].

In contrast, MRI enables noninvasive imaging with good spatial resolution of millimeters over a large brain volume, making it an appropriate tool for *in vivo* functional brain imaging. To date, the mainstream of functional MRI (fMRI) utilizes hemodynamic responses driven by brain activation, such as the blood oxygen level-dependent (BOLD) effect[14]. However, while the BOLD contrast mechanism reflects dynamic changes in neuronal activity through neurovascular coupling, it provides inherently indirect and relatively slow, hemodynamic-responsive information of brain function[15]. On the other hand, many studies have attempted to explore the possibility of using MRI to directly detect neuronal activity[16,17]. These studies utilized neuronal current-dependent signal phase shifts[18-20] and magnitude decay[21-24], the Lorentz effect[25], high temporal resolution[26,27], ghost artifacts[28], or cell swelling[29,30], while concerns about sensitivity exist[31-36].

In this study, we investigated the possibility of using MRI to detect membrane potential changes induced by modulating ionic solutions. Specifically, we explored how T_2 relaxation time and magnetization transfer (MT) correlate with membrane potential changes, both *in vitro* and *in vivo*. *In vitro* experiments utilized two homogeneous and electrically non-active cells, i.e., neuroblastoma (SH-SY5Y) and leukemia (Jurkat) cell lines, providing a controlled environment free from hemodynamic effects. This aided in accurately evaluating changes in MR parameters with membrane potential. *In vivo* experiments, conducted on a rat model with a craniotomy-exposed cortex, aimed to reproduce the *in vitro* findings.

Results

In vitro changes in MR parameters induced by membrane potential

A non-excitable neuroblastoma cell line, SH-SY5Y, was selected to investigate the relationship between MR parameters and membrane potential modulated by ionic solutions. After culturing SH-SY5Y cells, they were suspended in extracellular media with various potassium ion concentrations ($[K^+]$), while maintaining constant osmolarity by adjusting $[Na^+]$. As the control condition, $[K^+]$ = 4.2 mM was selected. For depolarization conditions, $[K^+]$ = 20, 40, and 80 mM were used. For hyperpolarization conditions, $[K^+]$ = 0.2 and 1 mM were used. The suspended cells were concentrated with centrifugation in an acrylic container and scanned in a 9.4 T preclinical MRI system. The imaging slice was positioned 0.5 mm below the pellet-extracellular media interface to ensure that signals were acquired from cell pellet. Under each condition, T_2 and MT parameters such as pool size ratio (PSR) and magnetization transfer rate (k_{mf}) were measured (**Fig. 1**). The PSR value represents the ratio of hydrogen protons in macromolecules and free water, and k_{mf} represents the magnetization transfer rate of hydrogen protons from macromolecules to free water. In addition, the membrane potential of SH-SY5Y cells in each condition was separately measured via patch clamp recording.

The changes of T_2, PSR, and k_{mf} in SH-SY5Y cells when the membrane potential (V_m) was modulated by varying $[K^+]$ are shown in **Figure 2**, alongside the actual V_m measured via patch clamp recordings. We conducted statistical analyses to assess the effect of changes in V_m from the control condition (ΔV_m) on the MR parameters. A linear mixed-effect model was applied to account for inter-sample variability and repeated measurements. This model included MR parameters as dependent variables, ΔV_m as a fixed effect, and cell batch as a random effect. The analysis yielded the following relationships:

$$T_2\ (\text{ms}) = 49.7\ (1 + 0.00687\ \Delta V_m) \cdots\cdots [1]$$

$$\text{PSR} = 0.0377\ (1 - 0.00542\ \Delta V_m) \cdots\cdots [2]$$

$$k_{mf}\ (\text{Hz}) = 14.8\ (1 + 0.000648\ \Delta V_m) \cdots\cdots [3]$$

The effects of ΔV_m on T_2 and PSR were both significant ($p < 0.0001$), indicating an increase in T_2 and a decrease in PSR during depolarization at high $[K^+]$, with the opposite trend during hyperpolarization at low $[K^+]$. On the other hand, the effect of ΔV_m on k_{mf} was not significant ($p = 0.360$).

Subsequent post-hoc analyses compared each experimental condition to the control using Dunnett's test to account for multiple comparisons (**Fig. 2**). During depolarization induced by the highest $[K^+]$ (80 mM, ΔV_m = 30.0 mV), changes in MR parameters were observed as a 20.0% increase in T_2 (ΔT_2 = 10.1 ms, $p < 0.0001$) and a 12.9% decrease in PSR (ΔPSR = −0.00476, $p < 0.0001$). Conversely, during hyperpolarization induced by the lowest $[K^+]$ (0.2 mM, ΔV_m = −5.33 mV), T_2 decreased by 4.40% (ΔT_2 = −2.21 ms, $p < 0.0001$) and PSR increased by 6.28% (ΔPSR = 0.00231, $p < 0.0005$). However, changes in k_{mf} were not significant across all conditions ($p > 0.05$). These findings from *in vitro* SH-SY5Y cell experiments suggest that MR parameters, such as T_2 and PSR, exhibit sufficient sensitivity to detect alterations in membrane potential induced by varying $[K^+]$, including both depolarization and hyperpolarization.

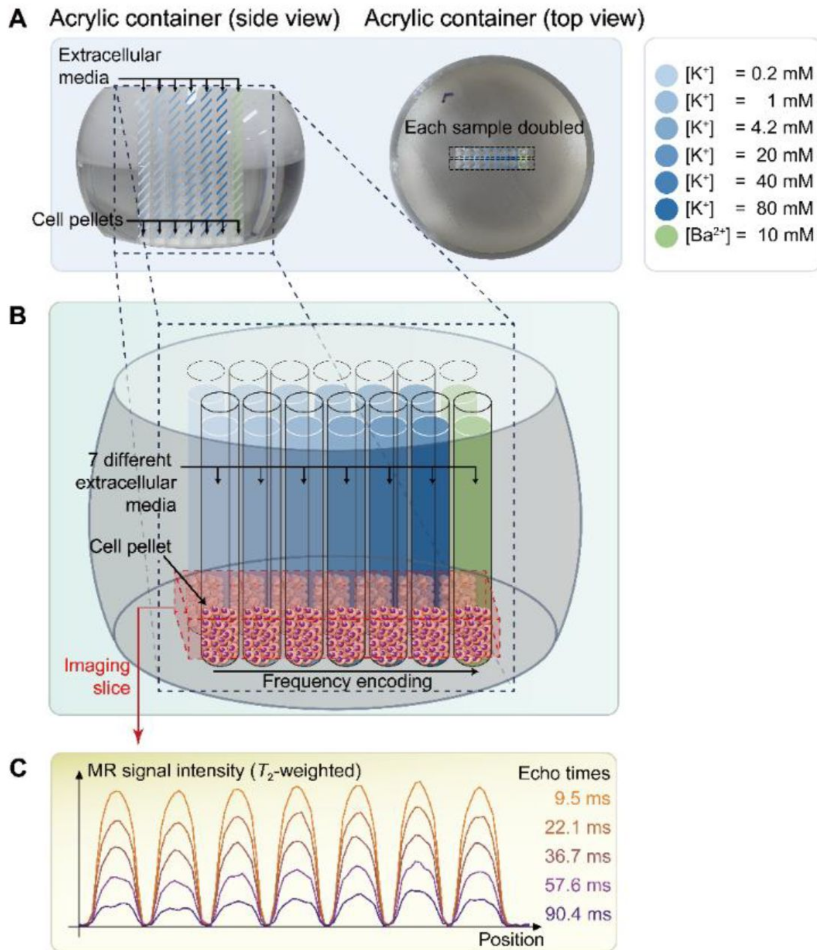


Fig. 1.

The schematic diagram of the *in vitro* experiment. (A) The picture on the left displays a side view of a double-sided cut spherical acrylic container with fabricated wells filled with extracellular media and cell pellets. As depicted in the top-view picture on the right, fourteen wells (matrix = 2 × 7) were created on the acrylic container, allowing each of the seven samples with six different K^+ concentrations ([K^+] = 0.2–80 mM) and one Ba^{2+} concentration ([Ba^{2+}] = 10 mM) to be doubled in the same column for improved signal-to-noise ratio (SNR) in MR signal acquisition. (B) The image illustrates the configuration after loading cells into the wells and pelleting them at the bottom of the wells. The imaging slice was positioned 0.5 mm below the pellet-media interface to acquire signals predominantly from the cell pellets. (C) Representative one-dimensional T_2-weighted MR signals with 5 selected echo times out of a total of 50 acquired echo times.

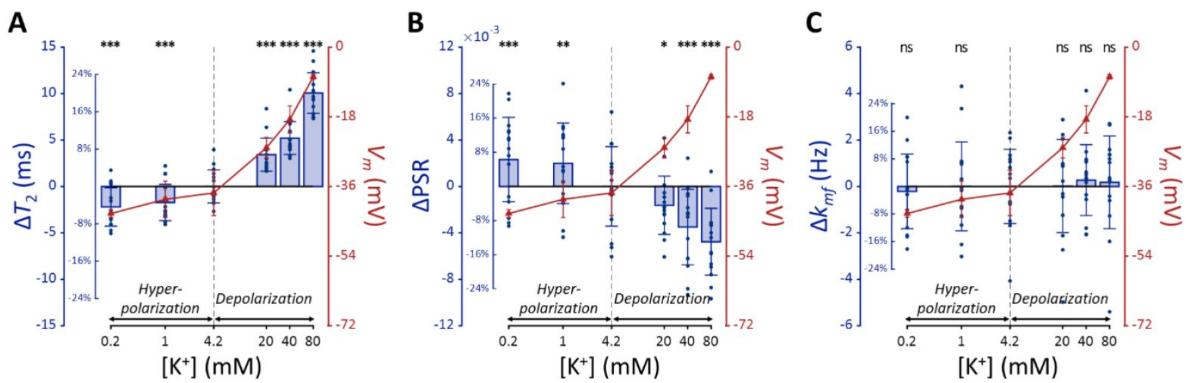


Fig. 2.

MR parameters and membrane potential (V_m) of SH-SY5Y cells versus extracellular K^+ concentrations ($[K^+]$). Changes in (A) T_2, (B) PSR, and (C) k_{mf} are displayed with blue bars (n = 15). Membrane potentials are plotted with red triangles (n = 3). The abscissa is in logarithmic scale. Error bars denote standard deviation. Statistical significance of changes in MR parameters is marked with asterisks (ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

Using a K^+ channel blocker

In this experiment, we investigated whether depolarization induced by altering potassium permeability with barium ions (Ba^{2+}) would affect MR parameters similarly to depolarization induced by varying [K^+], thereby further validating our findings. For this purpose, we administered barium ions (Ba^{2+}) at a concentration of 10 mM to induce depolarization while maintaining constant osmolarity by adjusting [Na^+]. Ba^{2+} was chosen because it inhibits several types of two-pore-domain potassium channels[37,38], which predominantly regulate the resting membrane potential. Previous studies have confirmed that Ba^{2+} depolarizes the membrane potential in SH-SY5Y and Jurkat cells[39,40].

The Ba^{2+}-induced depolarization condition was compared with K^+-induced depolarization and hyperpolarization conditions (**Fig. 3**). To compare the effects of [Ba^{2+}] and [K^+] on MR parameters, a linear mixed-effect model was utilized. This model included MR parameters as dependent variables, ΔV_m and its interaction with a group variable (indicating whether ΔV_m was induced by [Ba^{2+}] or [K^+]) as fixed effects, and cell batch as a random effect. This analysis revealed no significant interactions for all MR parameters assessed (p = 0.182 for T_2, p = 0.788 for PSR, and p = 0.0890 for k_{mf}). These findings suggest that changes in T_2 and PSR by membrane potential do not depend on the specific method of altering the membrane potential, whether by varying [K^+] or applying [Ba^{2+}]. This implies that if the changes in MR parameters observed with varying [K^+] were not primarily due to membrane potential but were due to unique K^+-related mechanism, then experiments using [Ba^{2+}] would not result in similar changes.

Subsequent post-hoc analyses compared the [Ba^{2+}]-induced depolarization to the control using Dunnett's test to account for multiple comparisons (**Fig. 3**). In response to depolarization caused by [Ba^{2+}] = 10 mM, T_2 increased by 7.82% (ΔT_2 = 3.93 ms, $p < 0.0001$) and PSR decreased by 8.06% (ΔPSR = −0.00297, $p < 0.0001$). The change in k_{mf} was not significant (Δk_{mf} = −0.832 Hz, p = 0.263). The depolarization of membrane potential induced by [Ba^{2+}] was measured as ΔV_m = 13.3 mV by patch clamp recording.

Using another cell type

To investigate whether membrane potential-modulating ionic solutions produce similar changes in MR parameters across different cell types, we assessed another cell line, Jurkat, under the same experimental conditions applied to SH-SY5Y cells. These conditions included a control condition ([K^+] = 4.2 mM), hyperpolarization under decreased [K^+] conditions ([K^+] = 0.2 and 1 mM), depolarization under increased [K^+] conditions ([K^+] = 20, 40, and 80 mM), and a Ba^{2+}-induced depolarization condition ([Ba^{2+}] = 10 mM), all maintaining consistent osmolarity by adjusting [Na^+].

Each experimental condition was compared to the control using Dunnett's test to account for multiple comparisons (**Fig. 4**). As observed with SH-SY5Y cells, Jurkat cells showed significant positive changes in T_2 and negative changes in PSR under increased [K^+] conditions. For example, at [K^+] = 80 mM, T_2 increased by 16.9% (ΔT_2 = 9.26 ms, $p < 0.0001$), PSR decreased by 21.9% (ΔPSR = −0.00347, $p < 0.01$). In contrast, during hyperpolarization at the lowest [K^+] = 0.2 mM, T_2 decreased by 18.1% (ΔT_2 = −9.93 ms, $p < 0.0001$), whereas PSR increased by 17.6% (ΔPSR = 0.00280, $p < 0.05$). The depolarization induced by [Ba^{2+}] = 10 mM resulted in a similar pattern, with T_2 increasing by 11.5% (ΔT_2 = 6.3 ms, $p < 0.0005$), although the decrease in PSR was not significant (ΔPSR = −0.00212, p = 0.211). Changes in k_{mf} remained non-significant across all conditions ($p > 0.05$). In summary, these findings indicate that detecting membrane potential changes induced by ionic solutions using MR parameters such as T_2 and PSR is not specific to a single cell type, although the magnitude of these changes may differ between cell types.

Fig. 3.

Changes in (A) T_2, (B) PSR, and (C) k_{mf} of SH-SY5Y cells across experimental conditions: $[K^+]$ = 0.2–80 mM (blue cross) and $[Ba^{2+}]$ = 10 mM (green triangle), compared to the control condition (black cross). Data from fifteen experiments (n = 15) are displayed. Linear regression lines for $[K^+]$ data (blue solid line) and $[Ba^{2+}]$ data (green solid line) are drawn along with dotted lines representing 95% confidence intervals. Error bars denote standard deviation. Statistical significance of changes in MR parameters with $[Ba^{2+}]$ = 10 mM is marked with asterisks (ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

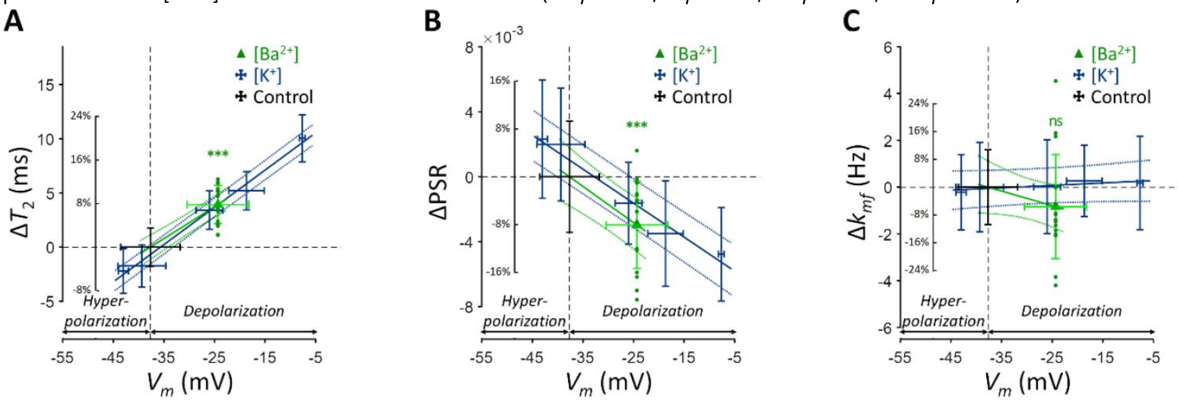
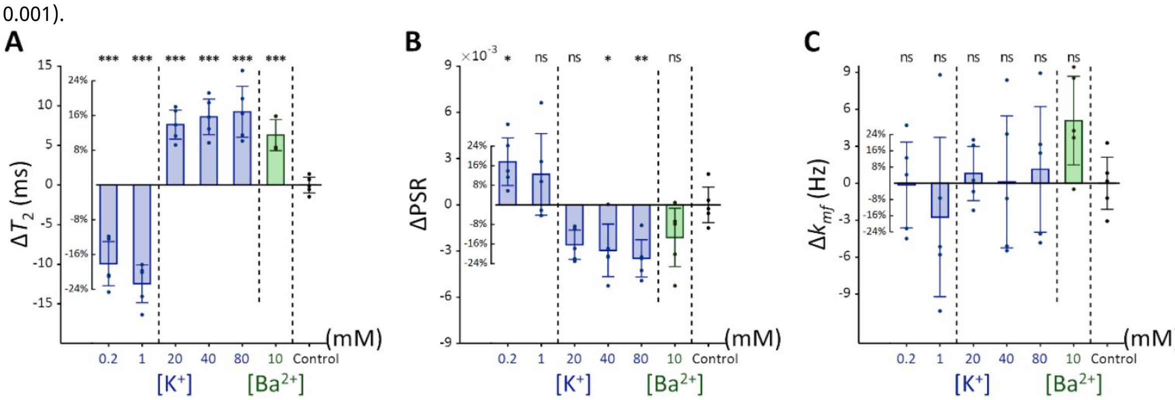


Fig. 4.

Changes in (A) T_2, (B) PSR, and (C) k_{mf} of Jurkat cells across experimental conditions: $[K^+]$ = 0.2–80 mM (blue bar) and $[Ba^{2+}]$ = 10 mM (green bar), compared to the control condition of $[K^+]$ = 4.2 mM (n = 5). Error bars denote standard deviation. Statistical significance of changes in MR parameters is marked with asterisks (ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).



In vivo changes in T_2 by membrane potential

The relationship of T_2 values and membrane potential modulated by [K^+], observed in the aforementioned SH-SY5Y and Jurkat cell studies, was further explored in an *in vivo* rat model to validate these findings under physiological conditions. As depicted in **Figure 5**, a craniotomy was performed to expose a 3-mm-diameter region of the rat cerebral cortex, followed by perfusion with artificial cerebrospinal fluid (aCSF) to modulate the membrane potential. Hemodynamic responses were pharmacologically suppressed. MRI scans were performed using a 7 T preclinical MRI system to measure T_2 in the exposed cortical area. A total of seven rats were used in the experiment that involved modulation of [K^+]. The experimental protocol included sequential application of four conditions, each lasting 12 minutes: a baseline condition at [K^+] = 3 mM, a depolarization condition at [K^+] = 40 mM, further depolarization at [K^+] = 80 mM, followed by a recovery condition using baseline aCSF. The recovery condition was applied to two of the seven rats. To distinguish the effect of aCSF perfusion on T_2 from the effect of changes in membrane potential, a control experiment was also conducted using only baseline aCSF for the entire duration (48 minutes) with another group of five rats.

In **Figure 6A**, a representative T_2 map is displayed with an enlarged image defining the ROI beneath the perfusion chamber (width = 1.8 mm, depth = 0.6mm). The average T_2 value within this ROI was estimated from the spatially averaged multi-echo spin-echo signal. A detailed analysis on the quality of these T_2 maps is presented in Supplementary Results S2.1. Changes in T_2 value (ΔT_2) were statistically analyzed using a linear mixed-effect model to account for inter-sample variability and repeated measurements. This model included ΔT_2 as a dependent variable, elapsed time and its interaction with the experiment type (i.e., [K^+]-modulation or control) as fixed effects, and a random effect for inter-sample variability.

Figure 6B shows the results of the statistical analysis. ΔT_2 values were plotted against elapsed time for [K^+]-modulation and control experiments. After 12 minutes from the initial condition, T_2 increased by 1.46% (ΔT_2 = 0.684 ms) in the [K^+]-modulation experiment ([K^+] = 40 mM) and by 0.223% (ΔT_2 = 0.104 ms) in the control experiment ([K^+] = 3 mM); the ΔT_2 difference (= 0.580 ms) between these experiments was not statistically significant (p = 0.0711). After 24 minutes, T_2 increased by 2.34% (ΔT_2 = 1.10 ms) in the [K^+]-modulation experiment ([K^+] = 80 mM) and by 0.386% (ΔT_2 = 0.181 ms) in the control experiment ([K^+] = 3 mM), with a significant difference in ΔT_2 (= 0.918 ms) between them (p = 0.00172). Due to the limited sample size in the recovery phase (n = 2 out of 7 for [K^+]-modulation), comparisons between experiments were not conducted for this recovery phase.

Our observations indicate that *in vivo* manipulation of membrane potential results in similar trends of T_2 changes as those observed *in vitro*, albeit with a smaller magnitude in the rat cortex. This discrepancy may be attributed to several factors: [K^+]-modulation affecting only a small portion of cells within the ROI; limited diffusion of aCSF through the leptomeninges; removal of excessive K^+ through clearing mechanisms; and differences in cell types (see the Discussion for more details).

Discussion

In this study, we demonstrated that MR parameters, specifically T_2 relaxation time and pool size ratio (PSR), can detect responses to membrane potential changes modulated by ionic solutions. Our *in vitro* experiments with cultured cells were designed to exclude physiological factors such as hemodynamic responses and respiration. We observed that depolarization increases T_2 and decreases PSR, while hyperpolarization has the opposite effect. *In vivo*, we pharmacologically

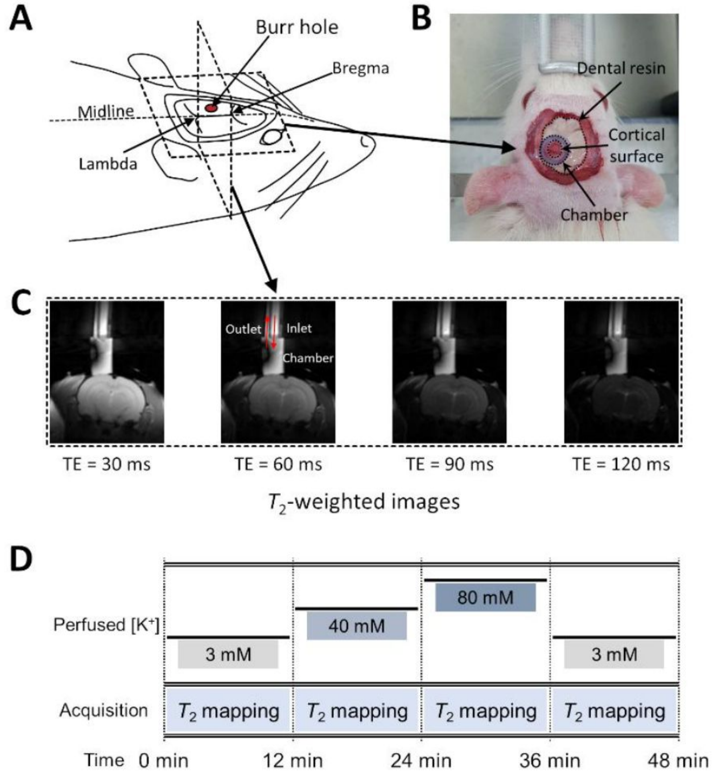


Fig. 5.

Experimental setup for *in vivo* manipulation of membrane potential. (A) A schematic diagram of the rat head post-craniotomy, showing the burr hole centered at 2.5 mm anterior and 2.0 mm lateral to the lambda. (B) Photograph of the rat head with a cylindrical chamber fixed over the burr hole, filled with artificial cerebrospinal fluid. (C) A representative series of T_2-weighted MR images for T_2 mapping. The chamber was connected to inlet and outlet perfusion tubes. (D) The experimental paradigm of the *in vivo* rat MR imaging. Four conditions were sequentially applied: control, depolarization, further depolarization, and recovery. Each condition lasted 12 minutes during which T_2 mapping were conducted.

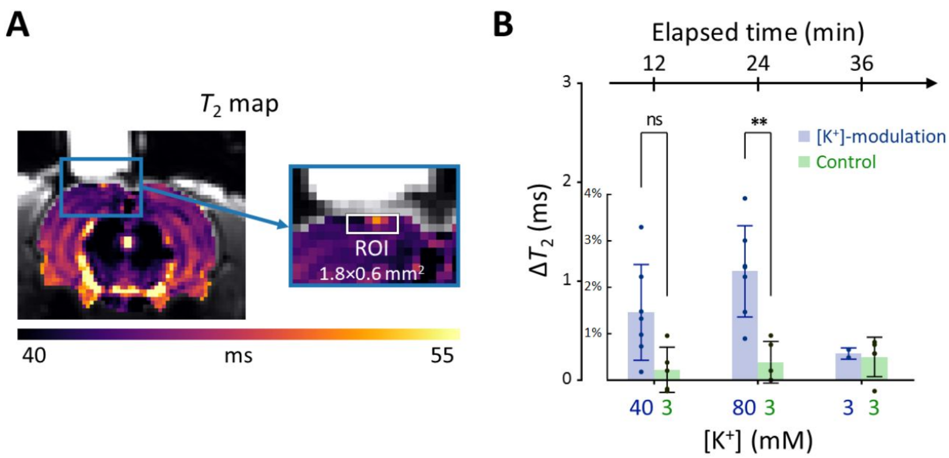


Fig. 6.

Results of the *in vivo* experiment results in rat models. (A) A representative T_2 map from a single rat with an enlarged image depicting the ROI for estimating average T_2 in the exposed cortical area, marked by a white rectangle (width = 1.8 mm, depth = 0.6 mm). (B) The changes in T_2 values within the ROI is plotted against elapsed time from the initial conditions. [K^+] in the perfused artificial cerebrospinal fluid are indicated on the bottom abscissa. Results from the [K^+]-modulation experiments are shown with blue bars, and those from the control experiments are shown with green bars. Statistical significance of the T_2 changes are marked with asterisks (ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$).

suppressed hemodynamic responses to minimize their impact on T_2 measurements. The trend of T_2 dependence on membrane potential *in vivo* was consistent with *in vitro* findings. However, the magnitude of T_2 changes in rat cortex was approximately one-ninth of that observed in SH-SY5Y cells *in vitro* at $[K^+]$ = 80 mM.

Other studies have also reported MR-detectable changes in response to extracellular $[K^+]$ modulation. For instance, research on spreading depression, which was induced by significantly high $[K^+]$ (~1 M) applied topically to rat cortex, has revealed detectable changes in T_1, T_2, and magnetization transfer ratio changes[41]; and spin-lock fMRI signals[42]. Similarly, increased $[K^+]$ has been studied *in vitro* with brain slices, linking cell volume changes to proton density-weighted MR signal changes[43]. T_2 mapping in Jurkat cells with increased [K+] conditions has also been investigated[27] and linked to cell volume change[44]. Our study complements these findings by employing direct membrane potential measurement via patch clamp, testing additional ionic agents such as Ba^{2+}, and demonstrating the phenomenon *in vivo*. Further reinforcement of these findings could be achieved by simultaneous recording of cell volume and other cellular characteristics to elucidate the underlying mechanisms more completely.

Interestingly, the MR responses in Jurkat cells differed from those in SH-SY5Y cells. While SH-SY5Y cells showed a near-linear dependence of MR parameters on log $[K^+]$, Jurkat cells displayed more step-like behavior. The reasons for this discrepancy remain unclear, but it suggests that the relationship between membrane potential and MR parameters may not strictly follow a simple log [K+] relationship.

Several factors may contribute to discrepancies between *in vivo* and *in vitro* results. For instance, the actual extracellular $[K^+]$ experienced by cells in the rat cortex may be lower than that of the perfused aCSF due to diffusion-limiting barriers such as the leptomeninges[45,46], even after the removal of the dura mater. Additionally, removal of excessive K^+ by clearing mechanisms[47,48] may further reduce the $[K^+]$ experienced by the cells. Partial volume effects and cell type differences may also contribute to the discrepancy.

From a biophysical perspective, the sensitivity of T_2 and PSR to membrane potential likely arises from alterations in cell volume, hydration water, and bulk water. Depolarization or hyperpolarization can lead to cell swelling or shrinking[49-51], influencing the proportion of cellular contents within the imaging voxel and thereby affecting MR parameters. Notably, neuronal or glial cell swelling has been proposed as a possible mechanism underlying diffusion fMRI[29,52,53] and previous MRI studies[43,44]. Although not as sensitive as T_2, our supplementary T_1 measurements (Supplementary Fig. S2–4) also exhibited similar trends in response to membrane potential changes. Moreover, hydration water, which has a significantly shorter T_2 than bulk water due to slower re-orientational and diffusive motions[54], may contribute to the observed MR changes. In particular, the correlation between PSR and membrane potential indicates that depolarization decreases the density of hydration water on the cell membrane within a voxel, thereby reducing PSR and simultaneously increasing T_2 due to a corresponding increase in free water, as well as cell swelling. Conversely, hyperpolarization may increase hydration water density, elevating PSR and lowering T_2. These interpretations are supported by recent optical studies showing reduced membrane hydration water during depolarization[55].

Several challenges and considerations in this study warrants discussion. First, maintaining the desired environment (37°C and 5% CO_2) for *in vitro* cells during MRI scans is challenging. In addition, intracellular accumulation of Ba^{2+} in the Ba^{2+}-induced depolarization experiment may affect cellular integrity. In this study, high cell viability (> 97.6%) was confirmed using cell viability assays under experimental conditions (Supplementary Fig. S5). Second, differences in T_2 value among the extracellular media may bias T_2 measurements due to the partial volume effect. However, in this study, the differences in T_2 among the extracellular media were found to be negligible compared to the observed T_2 changes (Supplementary Fig. S6). Third, while our findings show that membrane potential-modulating ionic solutions can affect MR parameters, it is important to note that these changes do not measure the membrane potential itself. Fourth, other factors such as pH, energy depletion, or extracellular osmolarity may affect MR parameters by altering cell volume. To minimize the effects of these other contributors, we matched osmolarity across all conditions, provided sufficient glucose to prevent energy depletion, and regulated pH levels by buffering with HEPES. Additionally, to mitigate possible changes in intra/extracellular volume fraction changes caused by cell movements, potentially due to agitation, we centrifuged the cells and imaged the bottom portion of the cell pellet, where cell movement was restricted due to close packing. Fifth, in the *in vivo* study, we attempted to suppress hemodynamic responses through pharmacological means, using a combination of N_{ω}-Nitro-L-arginine and nifedipine, both of which are known to inhibit hemodynamic responses in different ways[56,57], but their effects were not directly confirmed in this study. Future studies that simultaneously evaluate hemodynamic responses would strengthen our conclusions. Finally, our experimental paradigm was based on clamping the membrane potential at a specific level, thus measuring changes in T_2 and PSR during static depolarization or hyperpolarization rather than dynamic changes such as those seen during action potentials. Future research could explore temporally varying membrane potential to evaluate the dynamic correlation between membrane potential and MR parameters with high temporal resolution MRI[26,27].

In summary, our study demonstrates that MR parameters such as T_2 relaxation time can detect responses to membrane potential-modulating ionic solutions both *in vitro* and *in vivo*. This finding proposes a potential approach for noninvasively detecting changes in membrane potential using MRI.

Methods

In vitro cell culture

Two human cell lines were utilized for the experiments: SH-SY5Y, an immortalized neuroblastoma line, and Jurkat, a leukemia cell line. Both cell lines were sourced from the American Type Culture Collection (ATCC). The SH-SY5Y cells were cultured in DMEM/F12 medium supplemented with 10% (v/v) fetal bovine serum (FBS) and 100 U/ml penicillin/streptomycin. The cells were maintained at a constant temperature of 37°C in a humidified atmosphere containing 5% CO_2. Similarly, the Jurkat cells were cultured in RPMI-1640 medium, also supplemented with 10% (v/v) FBS and 100 U/ml penicillin/streptomycin, under the same conditions of temperature and CO_2 concentration.

In vitro manipulation of membrane potential with extracellular media

The baseline extracellular medium was prepared with the following components: KCl = 4.2 mM; NaCl = 145.8 mM; HEPES = 20 mM; glucose = 4.5 g/l; EGTA = 10 µM; pH = 7.2. This baseline medium was considered a control condition for various extracellular media used to adjust membrane potential. Two extracellular media with low K^+ concentrations (KCl = 0.2 and 1 mM) were prepared to hyperpolarize the membrane potential. Three extracellular media with high K^+ concentrations (KCl = 20, 40, and 80 mM) were prepared to depolarize the membrane potential. A Ba^{2+} medium containing 10 mM $BaCl_2$ was also prepared to depolarize the membrane potential in a different way, i.e., as a K^+ channel blocker. These seven extracellular media were used for both MR imaging and patch clamp recording *in vitro*. Sodium ion concentrations ([Na^+]) in all media were controlled to match the osmolarity with the baseline medium. The composition of all extracellular media is detailed in Supplementary Table S3.

Preparation of cells for *in vitro* MR measurement

SH-SY5Y cells were dissociated from their culture plates using 0.5 mM EDTA solution, then concentrated into a pellet (~70 μl) by centrifugation at 250 × g for two minutes. The pellet was resuspended in the culture medium and divided evenly into seven aliquots. Each cell suspension was centrifuged and resuspended in each of the seven different media specified in Supplementary Table S3. Centrifugation and resuspension were repeated three more times to completely clear out the culture medium. Each cell suspension with a different extracellular medium was then loaded into two wells on the same column of an acrylic container with 14 wells (matrix = 2 × 7) and centrifuged again to concentrate into pellets (**Fig. 1**). The acrylic container was purposely formed into a spherical segment to enhance the homogeneity of the static magnetic field[58]. The preparation of Jurkat cell samples was the same as for the SH-SY5Y cell samples. The preparation required 40–60 minutes, followed by an incubation period of 20–30 minutes.

In vitro MRI experiment

In vitro MRI experiments were performed on a 9.4 T MRI system (BioSpec 94/30 USR, Bruker BioSpin) at room temperature. A volume coil with an inner diameter of 86 mm was utilized for both radiofrequency (RF) pulse transmission and signal reception. Within the acrylic container, two wells on the same column (matrix = 2 × 7) contained identical cell pellets with the same extracellular media. MRI signals from seven different cell samples in the horizontal direction were separated by one-dimensional frequency encoding along that direction (**Fig. 1**). The MRI pulse sequence employed for mapping the T_2 value was a single-echo spin-echo (SESE) sequence with 50 variable echo times (TE) spaced between 9.5 and 290.5 ms on a logarithmic scale. For mapping the MT parameters, an inversion recovery multi-echo spin-echo (IR-MESE) sequence was used. The inversion times (TI) for the IR-MESE sequence were optimized using the theory of Cramér-Rao lower bounds[59]. The optimized TIs ranged from 4 to 10,079.4 ms. After each TI, 16 spin-echo trains were acquired with an echo spacing of 9.5 ms. Total scan time for both sequences was 23 minutes. Experiments were repeated 15 times for SH-SY5Y cells and 7 times for Jurkat cells, replacing cells in each repetition. Other scan parameters are detailed in Supplementary Table S1.

Animals

Male Wistar rats aged 8 weeks (250–300 g, Orient Bio) were used for MRI experiments after undergoing a craniotomy. All animal experiments were approved by the Institutional Animal Care and Use Committee at the National Cancer Center Korea (NCC-22-740). The rats were housed in ventilated cages under a 12h/12h light/dark cycle and provided with *ad libitum* access to food and water.

In vivo manipulation of membrane potential with artificial cerebrospinal fluid (aCSF)

The membrane potential of the exposed cortex was manipulated by directly perfusing the region-of-interest of the cerebral cortex with aCSF after a craniotomy. The baseline aCSF was prepared with the following components: KCl = 3 mM; NaCl = 135 mM; $MgCl_2$ = 3 mM; HEPES = 20 mM; glucose = 4.5 g/l; EGTA = 2 mM; N_ω-Nitro-L-arginine = 1 mM; Nifedipine = 0.1 mM; pH = 7.4. Hemodynamic effects were pharmacologically suppressed using N_ω-Nitro-L-arginine, Nifedipine, and EGTA. N_ω-Nitro-L-arginine suppresses depolarization-induced hemodynamic response by blocking the synthesis of nitric oxide, which acts as a vasodilator[56]. Nifedipine blocks voltage-sensitive Ca^{2+} channels, and EGTA chelates free Ca^{2+} to inhibit the hemodynamic response[57]. To induce depolarization, aCSF with high $[K^+]$ (KCl = 40 and 80 mM) was prepared. The osmolarity of the aCSF was matched with the concentration of NaCl.

Rat surgery

A craniotomy was performed on a Wistar rat. The experimental setup after a surgical procedure is illustrated in **Figure 5**. The surgery was performed following an established protocol[60] alike to that used in other MRI studies[41,42]. Anesthesia was induced with 3% isoflurane in O_2 and maintained with 2–3% isoflurane during the surgical procedure. Body temperature was maintained at 36.5–37.5°C with an infrared lamp. The head was fixed with a small animal stereotaxic frame. The hair on the scalp was shaved with a veterinary clipper. The skin and periosteum over the skull were removed using a scalpel and surgical scissors. A 3.0-mm-diameter burr hole was opened using a dental drill with its center at the coordinates of 2.5 mm anterior and 2.0 mm lateral to the lambda. Then, a cylindrical chamber was implanted upon the burr hole with cyanoacrylate glue and dental composite resin. The chamber was filled with the baseline aCSF and connected to inlet and outlet perfusion tubes. The exposed cerebral cortex inside the chamber was perfused with the baseline aCSF at a flow rate of 0.6 ml/min using peristaltic pumps.

In vivo MRI experiment

The rat with a cranial chamber installed on the cortical surface was placed on a 7 T MRI system (BioSpec 70/20 USR, Bruker BioSpin), fixed in a customized cradle with two ear-bars and a bite-bar. A customized surface coil (rectangular, 35 mm × 20 mm) was used for RF pulse transmission and signal reception. Body temperature was maintained at 36.5–37.5°C using a warm air blower. Anesthesia was maintained with 2% isoflurane in O_2 (0.6 l/min). Respiration rate and body temperature were monitored throughout the MRI experiment. MR images were acquired in a 2 mm coronal slice through the center of the burr hole (**Fig. 5**), using a multi-echo spin-echo (MESE) sequence with 20 TEs (7.5 to 150 ms). Other scan parameters are detailed in Supplementary Table S2.

A total of seven rats were subjected to four sequential experimental conditions, as depicted in **Figure 5**. First, the exposed cerebral cortex was perfused with baseline aCSF ([K^+] = 3 mM). Second, as a depolarizing condition, the perfusion media was switched to depolarizing aCSF of [K^+] = 40 mM. Third, membrane potential was further depolarized by perfusion with aCSF of [K^+] = 80 mM. Finally, as a recovery condition, the perfused aCSF was changed back to baseline aCSF. During each condition, MR images were acquired with MESE sequences for 12 minutes. Two rats underwent the whole four conditions, while other five rats did not undergo the recovery condition. As a control experiment, another set of rats (n = 5) underwent perfusion with the baseline aCSF for the same duration (48 minutes) as the previous experiment, and MR images were acquired with four MESE sequences, each for 12 minutes. Throughout the experiment, the perfusion rate was maintained at 0.6 ml/min.

Quantification of MR parameters

For the *in vivo* MR images acquired with a MESE sequence, echo trains were matched with a simulated dictionary of decay curves of multi-echo spin-echo signals created with the stimulated echo and slice profile correction[61] to estimate T_2 values. For the *in vitro* one-dimensional MR images acquired with the SESE sequence, signals were fitted to a mono-exponential function to estimate T_2 values. For the *in vitro* one-dimensional MR images acquired with the IR-MESE sequence, signals were fitted to a bi-exponential function[62,63] to estimate MT parameters. 16 spin echoes acquired after each TI were averaged to improve SNR. The MT parameters such as PSR and k_{mf} were derived from this fitting process. The detailed procedure for calculating T_2 and two MT parameters (i.e., PSR and k_{mf}) is provided in Supplementary Methods S1.1.

Patch clamp recordings

The membrane potential of SH-SY5Y cells was recorded at room temperature using the whole-cell mode of the patch clamp technique[64]. The bath solution was the same as the extracellular medium used in the MRI experiment and was constantly perfused at a flow rate of 2 ml/min. The composition of the pipette solution was as follows: KCl = 140 mM; NaCl = 5 mM; MgCl2 = 3 mM; HEPES = 10 mM; Mg-ATP = 1 mM; Na-GTP = 0.5 mM. The pH of the pipette solution was adjusted to 7.4 using KOH. Calcium ions (Ca^{2+}) were not included in the pipette solution to minimize Ca^{2+}-dependent currents. The resistance of the electrode was 3–5 MΩ with the internal solution filled. Recordings were performed with a patch amplifier (Axopatch-1D; Axon Instruments) and a current clamp was also used to monitor the membrane potential. The experiment was repeated three times, with cells replaced at each repetition.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Additional information

Author contributions

Conceptualization: K.M., S.-K.L., and J.-Y.P. Methodology: K.M., S.C., S.-K.L., J.L., P.T.T., D.K., J.S.L., and J.-Y.P. Investigation: K.M., S.C., P.T.T., and D.K.; Resources: S.C., J.L., D.K., J.S.L., and J.-Y.P. Visualization: K.M., and S.C. Supervision: J.L., and J.-Y.P. Writing – original draft K.M., and J.-Y.P. Writing – review & editing: K.M., S.C., S.-K.L., J.L., P.T.T., D.K., J.S.L., and J.- Y.P.

Additional files

Supplementary Information

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Author information

Kyeongseon Min

Department of Electrical and Computer Engineering, Seoul National University, Seoul, Republic of Korea

Sungkwon Chung

Department of Physiology, Sungkyunkwan University School of Medicine, Suwon, Republic of Korea

Seung-Kyun Lee

Department of Biomedical Engineering, Sungkyunkwan University, Suwon, Republic of Korea, Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University, Suwon, Republic of Korea

Jongho Lee
Department of Electrical and Computer Engineering, Seoul National University, Seoul, Republic of Korea

Phan Tan Toi
Department of Biomedical Engineering, Sungkyunkwan University, Suwon, Republic of Korea, Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University, Suwon, Republic of Korea
ORCID iD: 0000-0002-8073-2898

Daehong Kim
National Cancer Center, Goyang, Republic of Korea

Jung Seung Lee
Department of Biomedical Engineering, Sungkyunkwan University, Suwon, Republic of Korea, Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University, Suwon, Republic of Korea

Jang-Yeon Park
Department of Biomedical Engineering, Sungkyunkwan University, Suwon, Republic of Korea, Department of Intelligent Precision Healthcare Convergence, Sungkyunkwan University, Suwon, Republic of Korea

For correspondence: jyparu@skku.edu

Editors

Reviewing Editor
Timothy Behrens
University of Oxford, Oxford, United Kingdom

Senior Editor
Tamar Makin
University of Cambridge, Cambridge, United Kingdom

Reviewer #1 (Public review):

Summary:

This paper examines changes in relaxation time (T1 and T2) and magnetization transfer parameters that occur in a model system and in vivo when cells or tissue are depolarized using an equimolar extracellular solution with different concentrations of the depolarizing ion K+. The motivation has been revised to state that the results suggest a potential approach to non-invasively detect changes in membrane potential using MRI.

Strengths:

The authors argue that the use of various concentrations of KCL in the extracellular fluid depolarize or hyperpolarize the cell pellets used, and that this change in membrane potential is the driving force for the T2 (and T1-supplementary material) changes observed. In particular, they report an increase in T2 with increasing KCL concentration in the extracellular fluid (ECF) of pellets of SH-SY5Y cells. To offset the increasing osmolarity of the ECF due to the increase in KCL, the NaCL molarity of the ECF is proportionally reduced. The authors measure the intracellular voltage using patch clamp recordings, which is a gold standard. With 80 mM of KCL in the ECF, a change in T2 of the cell pellets of ~10 ms is observed with the intracellular potential recorded as about -6 mv. A very large T1 increase of ~90 ms is reported under the same conditions. The PSR (ratio of hydrogen protons on macromolecules to free water) decreases by about 10% at this 80 mM KCL concentration. Similar results are seen in a Jurkat cell line and similar, but far smaller changes are observed in vivo, for a variety of reasons discussed. As a final control, T1 and T2 values are measured in the various equimolar KCL solutions. As expected, no significant changes in T1 and T2 of the ECF were observed for these concentrations.

Weaknesses:

While the concepts presented are interesting, and the actual experimental methods seem to be nicely executed, the conclusions are not supported by the data for a number of reasons. This is not to say that the data isn't consistent with the conclusions, but there are other controls not included that would be necessary to draw the conclusion that it is membrane potential that is driving these T1 and T2 changes. The results are consistent with Stroman et al. Magn. Reson. in Med. 59:700-706 (increased T2 with KCL) as well as some other cited work. However all those authors emphasize that cell swelling is the mechanism, not cell membrane potentials.

It is well established that cells swell/shrink upon depolarization/hyperpolarization. Cell swelling is accompanied by increased light transmittance in vivo, and this should be true in the pellet system as well. In a beautiful series of experiments, Stroman et al. (2008) showed in perfused brain slices that the cells swell upon equimolar KCL depolarization and the light transmittance increases. The time course of these changes is quite slow, of the order of many minutes, both for the T2-weighted MRI signal and for the light transmittance. Stroman et al. also show that hypoosmotic changes produce the exact same timecourse as the KCL depolarization changes (and vice versa for the hyperosmotic changes - which cause cell shrinkage). Their conclusion therefore, was that cell swelling (not membrane potential) was the cause of the T2-weighted changes observed, and that these were relatively slow (on the scale of many minutes).

What are the implications for the current study? Well, for one, the authors cannot exclude cell swelling as the mechanism for T2 changes, as they have not measured that. It is however well established that cell swelling occurs during depolarization, so this is not in question. Water in the pelletized cells is in slow/intermediate exchange with the ECF, and the solutions for the two compartment relaxation model for this are well established (see Menon and Allen, Magn. Reson. in Med. 20:214-227 (1991). The T2 relaxation times should be multiexponential (see point (3) further below). The current work cannot exclude cell swelling as the mechanism for T2 changes (it is mentioned in the paper, but not dealt with). Water entering cells dilutes the protein structures, changes rotational correlation times of the proteins in the cell and is known to increase T2. The PSR confirms that this is indeed happening, so the data in this work is completely consistent with the Stroman work and completely consistent with cell swelling associated with depolarization. The authors should have performed light scattering studies to demonstrate the degree cell swelling or shrinkage. Measuring intracellular potential is not enough to clarify the mechanism.

So why does it matter whether the mechanism is cell swelling or membrane potential? The reason is response time. Cell swelling due to depolarization is a slow process, slower than hemodynamic responses that characterize BOLD. And in fact, cell swelling under normal homeostatic conditions in vivo is virtually non-existent. Only sustained depolarization events typically associated with non-naturalistic stimuli or brain dysfunction produce cell swelling. Membrane potential changes associated with neural activity, on the other hand, are very fast. In this manuscript, the authors have convincingly shown a signal change that is virtually the same as what was seem in the Stroman publication, but they have not shown that there is a response that can be detected with anything approaching the timescale of an action potential. So one cannot definitely say that the changes observed are due to membrane potential. One can only say they are consistent with cell swelling, regardless of what causes the cell swelling. The First line of the discussion still claims that T2 relaxation time and pool size ratio (PSR) can detect responses to membrane potential changes modulated by ionic solutions. However, in the absence of cell swelling controls, this cannot be stated.

For this mechanism to be relevant to measuring neuronal activity directly or explaining techniques such DIANA, one needs to show that the cell swelling changes occur within a millisecond, which has never been reported. If one knows the populations of ECF and pellet, the T2s of the ECF and pellet and the volume change of the cells in the pellet, one can model any expected T2 changes due to neuronal activity. I think one would find that these are minuscule within the context of an action potential, or even bulk action potentials.

Comments on revisions:

The manuscript is well written and my previous methodological concerns have been clarified as well. There are no flaws in the experiments, but the interpretation really depends on simultaneous measurements of cell volume and membrane potential, which have yet to be done.

https://doi.org/10.7554/eLife.101642.2.sa2

Reviewer #2 (Public review):

Summary:

Min et al. attempt to demonstrate a mechanism whereby magnetic resonance imaging (MRI) can reflect changes in neuronal membrane potentials. They approach this goal by studying how MRI contrast and cellular potentials together respond to treatment of cultured cells with ionic solutions that are known to depolarize or hyperpolarize excitable cells. The authors specifically examine two MRI-based measurements: (A) the transverse (T2) relaxation rate, which reflects microscopic magnetic fields caused by solutes and biological structures; and (B) the fraction or "pool size ratio" (PSR) of water molecules estimated to be bound to macromolecules, using an MRI technique called magnetization transfer (MT) imaging. They see that depolarizing K+ and Ba2+ concentrations lead to T2 increases and PSR decreases that vary approximately linearly with parallel measurements of voltage in a neuroblastoma cell line and that change similarly in a second cell type. They also show that depolarizing potassium concentrations evoke T2 increases in rat brains, and that these changes are reversed when potassium is renormalized. Min et al. argue that their results suggest a basis for noninvasive functional imaging of cellular voltage signals. If this were true, it would help validate a recent paper published by some of the authors (Toi et al., Science 378:160-8, 2022), in which they claimed to be able to detect millisecond-scale neuronal responses by MRI.

Strengths:

The discovery of a mechanism for relating cellular membrane potential to MRI contrast could yield an important means for studying functions of the nervous system. Achieving this has been a longstanding goal in the MRI community, but previous strategies have proven insufficient for neuroscientific or clinical applications. The current paper suggests that one of the simplest and most widely used MRI contrast mechanisms-T2 weighted imaging-may indicate correlates of membrane potential if measured in the absence of the hemodynamic signals that most functional MRI (fMRI) experiments rely on. The authors make their case using quantitative tests that include some controls for ion and cell type-specificity of their in vitro results and reversibility of MRI changes observed in vivo.

Weaknesses:

The major weakness of the paper is that it uses only slow correlational experiments to probe the relationship between MRI contrast and membrane potential. The authors do not examine effects on the subsecond time scale that is of greatest interest, and they do not adequately consider how biophysical factors with only loose relationship to electrophysiological variables could explain their imaging results. Notably, depolarizing ionic solutions that perturb membrane potential can also induce changes in cellular volume and tissue structure that in turn alter MRI contrast properties similarly to the results shown here. For example, a study by Stroman et al. (Magn Reson Med 59:700-6, 2008) reported reversible potassium-dependent T2 increases in neural tissue that correlate closely with light scattering-based indications of cell swelling. Phi Van et al. (Sci Adv 10:eadl2034, 2024) showed that potassium addition to one of the cell lines used here likewise leads to cell size increases and T2 increases. In their revised manuscript, the authors acknowledge that cell swelling might contribute to the MRI signals they report, but they do nothing to probe the contributions or characteristics of such effects. If cell swelling accounted for the author's MRI results, it would likely operate on a time scale far too slow to yield useful indications of membrane potential. Given these considerations and the absence of data demonstrating correspondence of electrophysiological measures with MRI readouts on a fast time scale, the paper fails to provide evidence that membrane potential changes can be meaningfully detected by MRI.

https://doi.org/10.7554/eLife.101642.2.sa1

Author response:

The following is the authors' response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *This paper examines changes in relaxation time (T1 and T2) and magnetization transfer parameters that occur in a model system and in vivo when cells or tissue are depolarized using an equimolar extracellular solution with different concentrations of the depolarizing ion K^+. The motivation is to explain T2 changes that have previously been observed by the authors in an in vivo model with neural stimulation (DIANA) and to try to provide a mechanism to explain those changes.*
>
> *Strengths:*
>
> *The authors argue that the use of various concentrations of KCL in the extracellular fluid depolarize or hyperpolarize the cell pellets used and that this change in membrane potential is the driving force for the T2 (and T1-supplementary material) changes observed. In particular, they report an increase in T2 with increasing KCL concentration in the extracellular fluid (ECF) of pellets of SH-SY5Y cells. To offset the increasing osmolarity of the ECF due to the increase in KCL, the NaCL molarity of the ECF is proportionally reduced. The authors measure the intracellular voltage using patch clamp recordings, which is a gold standard. With 80 mM of KCL in the ECF, a change in T2 of the cell pellets of ~10 ms is observed with the intracellular potential recorded as about -6 mv. A very large T1 increase of ~90 ms is reported under the same conditions. The PSR (ratio of hydrogen protons on macromolecules to free water) decreases by about 10% at this 80 mM KCL concentration. Similar results are seen in a Jurkat cell line and similar, but far smaller changes are observed in vivo, for a variety of reasons discussed. As a final*

> *control, T1 and T2 values are measured in the various equimolar KCL solutions. As expected, no significant changes in T1 and T2 of the ECF were observed for these concentrations.*
>
> *Weaknesses:*
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> *[Reviewer 1, Comment 1] While the concepts presented are interesting, and the actual experimental methods seem to be nicely executed, the conclusions are not supported by the data for a number of reasons. This is not to say that the data isn't consistent with the conclusions, but there are other controls not included that would be necessary to draw the conclusion that it is membrane potential that is driving these T1 and T2 changes. Unfortunately for these authors, similar experiments conducted in 2008 (Stroman et al. Magn. Reson. in Med. 59:700-706) found similar results (increased T2 with KCL) but with a different mechanism, that they provide definite proof for. This study was not referenced in the current work.*
>
> *It is well established that cells swell/shrink upon depolarization/hyperpolarization. Cell swelling is accompanied by increased light transmittance in vivo, and this should be true in the pellet system as well. In a beautiful series of experiments, Stroman et al. (2008) showed in perfused brain slices that the cells swell upon equimolar KCL depolarization and the light transmittance increases. The time course of these changes is quite slow, of the order of many minutes, both for the T2-weighted MRI signal and for the light transmittance. Stroman et al. also show that hypoosmotic changes produce the exact same time course as the KCL depolarization changes (and vice versa for the hyperosmotic changes - which cause cell shrinkage). Their conclusion, therefore, was that cell swelling (not membrane potential) was the cause of the T2-weighted changes observed, and that these were relatively slow (on the scale of many minutes).*
>
> *What are the implications for the current study? Well, for one, the authors cannot exclude cell swelling as the mechanism for T2 changes, as they have not measured that. It is however well established that cell swelling occurs during depolarization, so this is not in question. Water in the pelletized cells is in slow/intermediate exchange with the ECF, and the solutions for the two compartment relaxation model for this are well established (see Menon and Allen, Magn. Reson. in Med. 20:214-227 (1991). The T2 relaxation times should be multiexponential (see point (3) further below). The current work cannot exclude cell swelling as the mechanism for T2 changes (it is mentioned in the paper, but not dealt with). Water entering cells dilutes the protein structures, changes rotational correlation times of the proteins in the cell and is known to increase T2. The PSR confirms that this is indeed happening, so the data in this work is completely consistent with the Stroman work and completely consistent with cell swelling associated with depolarization. The authors should have performed light scattering studies to demonstrate the presence or absence of cell swelling. Measuring intracellular potential is not enough to clarify the mechanism.*

[Reviewer 1, Response 1] We appreciate the reviewer's comments. We agree that changes in cell volume due to depolarization and hyperpolarization significantly contribute to the observed changes in T2, PSR, and T1, especially in pelletized cells. For this reason, we already noted in the Discussion section of the original manuscript that cell volume changes influence the observed MR parameter changes, though this study did not present the magnitude of the cell volume changes. In this regard, we thank the reviewer for introducing the work by Stroman et al. (Magn Reson Med 59:700-706, 2008). When discussing the contribution of the cell volume changes to the observed MR parameter changes, we additionally discussed the work of Stroman et al. in the revised manuscript.

In addition, we acknowledge that the title and main conclusion of the original manuscript may be misleading, as we did not separately consider the effect of cell volume changes on MR parameters. To more accurately reflect the scope and results of this study and also take into account the reviewer 2's suggestion, we adjusted the title to "Responses to membrane potential-modulating ionic solutions measured by magnetic resonance imaging of cultured cells and in vivo rat cortex" and also revised the relevant phrases in the main text.

Finally, when [K^+]-induced membrane potential changes are involved, there seems to be factors other than cell volume changes that appear to influence T^2 changes. Our follow-up study shows that there are differences in volume changes for the same T^2 change in the following two different situations: pure osmotic volume changes versus [K^+]-induced volume changes. For example, for the same T^2 change, the volume change for depolarization is greater than the volume change for hypoosmotic conditions. We will present these results in this coming ISMRM 2025 and are also preparing a manuscript to report shortly.

> *[Reviewer 1, Comment 2] So why does it matter whether the mechanism is cell swelling or membrane potential? The reason is response time. Cell swelling due to depolarization is a slow process, slower than hemodynamic responses that characterize BOLD. In fact, cell swelling under normal homeostatic conditions in vivo is virtually non-existent. Only sustained depolarization events typically associated with non-naturalistic stimuli or brain dysfunction produce cell swelling. Membrane potential changes associated with neural activity, on the other hand, are very fast. In this manuscript, the authors have convincingly shown a signal change that is virtually the same as what was seen in the Stroman publication, but they have not shown that there is a response that can be detected with anything approaching the timescale of an action potential. So one cannot definitely say that the changes observed are due to membrane potential. One can only say they are consistent with cell swelling, regardless of what causes the cell swelling.*
>
> *For this mechanism to be relevant to explaining DIANA, one needs to show that the cell swelling changes occur within a millisecond, which has never been reported. If one knows the populations of ECF and pellet, the T2s of the ECF and pellet and the volume change of the cells in the pellet, one can model any expected T2 changes due to neuronal activity. I think one would find that these are minuscule within the context of an action potential, or even bulk action potential.*

[Reviewer 1, Response 2] In the context of cell swelling occurring at rapid response times, if we define cell swelling simply as an "increase in cell volume," there are several studies reporting transient structural (or volumetric) changes (e.g., ~nm diameter change over ~ms duration) in neuron cells during action potential propagation (Akkin et al., Biophys J 93:1347-1353, 2007; Kim et al., Biophys J 92:3122-3129, 2007; Lee et al., IEEE Trans Biomed Eng 58:3000-3003, 2011; Wnek et al., J Polym Sci Part B: Polym Phys 54:7-14, 2015; Yang et al., ACS Nano 12:4186-4193, 2018). These studies show a good correlation between membrane potential changes and cell volume changes (even if very small) at the cellular level within milliseconds.

As mentioned in the Response 1 above, this study does not address rapid dynamic membrane potential changes on the millisecond scale, which we explicitly mentioned as one of the limitations in the Discussion section of the original manuscript. For this reason, we do not claim in this study that we provide the reader with definitive answers about the mechanisms involved in DIANA. Rather, as a first step toward addressing the mechanism of DIANA, this study confirms that there is a good correlation between changes in membrane potential and measurable MR parameters (e.g., T^2 and PSR) when using ionic solutions that modulate membrane potential. Identifying MR parameter changes that occur during millisecond-scale membrane potential changes due to rapid neural activation will be addressed in the follow-up study mentioned in the Response 1 above.

> *There are a few smaller issues that should be addressed.*

> *[Reviewer 1, Comment 3] (1) Why were complicated imaging sequences used to measure T1 and T2? On a Bruker system it should be possible to do very simple acquisitions with hard pulses (which will not need dictionaries and such to get quantitative numbers). Of course, this can only be done sample by sample and would take longer, but it avoids a lot of complication to correct the RF pulses used for imaging, which leads me to the 2nd point.*

[Reviewer 1, Response 3] We appreciate the reviewer's suggestion regarding imaging sequences. In fact, we used dictionaries for fitting in vivo T^2 decay data, not in vitro data. Sample-by-sample nonlocalized acquisition with hard pulses may be applicable for in vitro measurements. However, for in vivo measurements, a slice-selective multi-echo spin-echo sequence was necessary to acquire T^2 maps within a reasonable scan time. Our choice of imaging sequence was guided by the need to spatially resolve MR signals from specific regions of interest while balancing scan time constraints.

> *[Reviewer 1, Comment 4] (2) Figure S1 (H) is unlike any exponential T2 decay I have seen in almost 40 years of making T2 measurements. The strange plateau at the beginning and the bump around TE = 25 ms are odd. These could just be noise, but the fitted curve exactly reproduces these features. A monoexponential T2 decay cannot, by definition, produce a fit shaped like this.*

[Reviewer 1, Response 4] The T^2 decay curves in Figure S1(H) indeed display features that deviate from a simple monoexponential decay. In our in vivo experiments, we used a multi-echo spin-echo sequence with slice-selective excitation and refocusing pulses. In such sequences, the echo train is influenced by stimulated echoes and imperfect slice profiles. This phenomenon is inherent to the pulse sequence rather than being artifacts or fitting errors (Hennig, Concepts Magn Reson 3:125-143, 1991; Lebel and Wilman, Magn Reson Med 64:1005-1014, 2010; McPhee and Wilman, Magn Reson Med 77:2057-2065, 2017). Therefore, we fitted the T_2 decay curve using the technique developed by McPhee and Wilman (2017).

> *[Reviewer 1, Comment 5] (3) As noted earlier, layered samples produce biexponential T2 decays and monoexponential T1 decays. I don't quite see how this was accounted for in the fitting of the data from the pellet preparations. I realize that these are spatially resolved measurements, but the imaging slice shown seems to be at the boundary of the pellet and the extracellular media and there definitely should be a biexponential water proton decay curve. Only 5 echo times were used, so this is part of the problem, but it does mean that the T2 reported is a population fraction weighted average of the T2 in the two compartments.*

[Reviewer 1, Response 5] We understand the reviewer's concern regarding potential biexponential decay due to the presence of different compartments. In our experiments, we carefully positioned the imaging slice sufficiently remote from the pellet-media interface. This approach ensures that the signal predominantly arises from the cells (and interstitial fluid), excluding the influence of extracellular media above the cell pellet. We described the imaging slice more clearly in the revised manuscript. As mentioned in our Methods section, for in vitro experiments, we repeated a single-echo spin-echo sequence with 50 difference echo times. While Figure 1C illustrates data from five echo times for visual clarity, the full dataset with all 50 echo times was used for fitting. We clarified this point in the revised manuscript to avoid any misunderstanding.

> *[Reviewer 1, Comment 6] (4) Delta T1 and T2 values are presented for the pellets in wells, but no absolute values are presented for either the pellets or the KCL solutions that I could find.*

[Reviewer 1, Response 6] As requested by the reviewer, we included the absolute values in the supplementary information.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *Min et al. attempt to demonstrate that magnetic resonance imaging (MRI) can detect changes in neuronal membrane potentials. They approach this goal by studying how MRI contrast and cellular potentials together respond to treatment of cultured cells with ionic solutions. The authors specifically study two MRI-based measurements: (A) the transverse (T2) relaxation rate, which reflects microscopic magnetic fields caused by solutes and biological structures; and (B) the fraction or "pool size ratio" (PSR) of water molecules estimated to be bound to macromolecules, using an MRI technique called magnetization transfer (MT) imaging. They see that depolarizing K^+ and Ba2+ concentrations lead to T2 increases and PSR decreases that vary approximately linearly with voltage in a neuroblastoma cell line and that change similarly in a second cell type. They also show that depolarizing potassium concentrations evoke reversible T2 increases in rat brains and that these changes are reversed when potassium is renormalized. Min et al. argue that this implies that membrane potential changes cause the MRI effects, providing a potential basis for detecting cellular voltages by noninvasive imaging. If this were true, it would help validate a recent paper published by some of the authors (Toi et al., Science 378:160-8, 2022), in which they claimed to be able to detect millisecond-scale neuronal responses by MRI.*
>
> *Strengths:*
>
> *The discovery of a mechanism for relating cellular membrane potential to MRI contrast could yield an important means for studying functions of the nervous system. Achieving this has been a longstanding goal in the MRI community, but previous strategies have proven too weak or insufficiently reproducible for neuroscientific or clinical applications. The current paper suggests remarkably that one of the simplest and most widely used MRI contrast mechanisms-T2 weighted imaging-may indicate membrane potentials if measured in the absence of the hemodynamic signals that most functional MRI (fMRI) experiments rely on. The authors make their case using a diverse set of quantitative tests that include controls for ion and cell type-specificity of their in vitro results and reversibility of MRI changes observed in vivo.*
>
> *Weaknesses:*
>
> *[Reviewer 2, Comment 1] The major weakness of the paper is that it uses correlational data to conclude that there is a causational relationship between membrane potential and MRI contrast. Alternative explanations that could explain the authors' findings are not adequately considered. Most notably, depolarizing ionic solutions can also induce changes in cellular volume and tissue structure that in turn alter MRI contrast properties similarly to the results shown here. For example, a study by Stroman et al. (Magn Reson Med 59:700-6, 2008) reported reversible potassium-dependent T2 increases in neural tissue that correlate closely with light scattering-based indications of cell swelling. Phi Van et al. (Sci Adv 10:eadl2034, 2024) showed that potassium addition to one of the cell lines used here likewise leads to cell size increases and T2 increases. Such effects could in principle account for Min et al.'s results, and indeed it is difficult to see how they would not contribute, but they occur on a time scale far too slow to yield useful indications of membrane potential. The authors' observation that PSR correlates negatively with T2 in their experiments is also consistent with this explanation, given the inverse relationship usually observed (and mechanistically expected) between these two parameters. If the*

> *authors could show a tight correspondence between millisecond-scale membrane potential changes and MRI contrast, their argument for a causal connection or a useful correlational relationship between membrane potential and image contrast would be much stronger. As it is, however, the article does not succeed in demonstrating that membrane potential changes can be detected by MRI.*

[Reviewer 2, Response 1] We appreciate the reviewer's comments. We agree that changes in cell volume due to depolarization and hyperpolarization significantly contribute to the observed MR parameter changes. For this reason, we have already noted in the Discussion section of the original manuscript that cell volume changes influence the observed MR parameter changes. In this regard, we thank the reviewer for introducing the work by Stroman et al. (Magn Reson Med 59:700-706, 2008) and Phi Van et al. (Sci Adv 10:eadl2034, 2024). When discussing the contribution of the cell volume changes to the observed MR parameter changes, we additionally discussed both work of Stroman et al. and Phi Van et al. in the revised manuscript.

In addition, this study does not address rapid dynamic membrane potential changes on the millisecond scale, which we explicitly discussed as one of the limitations of this study in the Discussion section of the original manuscript. For this reason, we do not claim in this study that we provide the reader with definitive answers about the mechanisms involved in DIANA. Rather, as a first step toward addressing the mechanism of DIANA, this study confirms that there is a good correlation between changes in membrane potential and measurable MR parameters (although on a slow time scale) when using ionic solutions that modulate membrane potential. Identifying MR parameter changes that occur during millisecond-scale membrane potential changes due to rapid neural activation will be addressed in the follow-up study mentioned in the Response 1 to Reviewer 1's Comment 1 above.

Together, we acknowledge that the title and main conclusion of the original manuscript may be misleading. To more accurately reflect the scope and results of this study and also consider the reviewer's suggestion, we adjusted the title to "Responses to membrane potential-modulating ionic solutions measured by magnetic resonance imaging of cultured cells and in vivo rat cortex" and also revised the relevant phrases in the main text.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *[Reviewer 1, Comment 7] The manuscript is well written. One thing to emphasize early on is that the KCL depolarization is done in an equimolar (or isotonic) manner. I was not clear on this point until I got to the very end of the methods. This is a strength of the paper and should be presented earlier.*

[Reviewer 1, Response 7] In response to the reviewer's suggestion, we have revised the manuscript to present the equimolar characteristic of our experiment earlier.

> *[Reviewer 1, Comment 8] In terms of experiments, the relaxation time measurements are not well constructed. They should be done with a CPMG sequence with hundreds of echos and properly curve fit. This is entirely possible on a Bruker spectrometer.*

[Reviewer 1, Response 8] As noted in our Response to Reviewer 1's Comment 3, while a CPMG sequence with numerous echoes and straightforward curve fitting can be effective, it is less feasible for in vivo experiments. Our multi-echo spin-echo sequence was a balanced approach between spatial resolution, reasonable scan duration, and the need to localize signals within specific regions of interest.

> *[Reviewer 1, Comment 9] Measurements of cell swelling should be done to determine the time course of the cell swelling. This could be with NMR (CPMG) or with light scattering. For this mechanism to be relevant to explaining DIANA, one needs to show that the cell swelling changes occur within a millisecond, which has never been reported. If one knows the populations of ECF and pellet, the T2s of the ECF and pellet and the volume change of the cells in the pellet, one can model any expected T2 changes due to neuronal activity.*

[Reviewer 1, Response 9] We acknowledge the importance of further research to further strengthened the claims of this study through additional experiments such as cell volume recording. We will do it in future studies.

As noted in our Response 2 to Reviewer 1’s Comment 2, this study does not address rapid membrane potential changes on the millisecond scale, and we acknowledge that establishing the precise timing of cell swelling is crucial for fully understanding the mechanisms of DIANA. Our current work demonstrates that MR parameters (e.g., T^2 and PSR) correlate strongly with membrane potential-modulating ionic environments, but it does not extend to millisecond-scale neural activation. We recognize the importance of further experiments, such as direct cell volume measurements and plan to incorporate it in future studies to build on the insights gained from the present work.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *Here are a few comments, questions, and suggestions for improvement:*
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> *[Reviewer 2, Comment 2] I could not find much information about the various incubation times and delays used for the authors' in vitro experiments. For each of the in vitro experiments in particular, how long were cells exposed to the stated ionic condition prior to imaging, and how long did the imaging take? Could this and any other relevant information about the experimental timing please be provided and added to the methods section?*

[Reviewer 2, Response 2] We have included the information about the preparation/incubation times in the revised manuscript. For the scan time, it was already stated in the original manuscript: 23 minutes for the single-echo spin-echo sequence and 23 minutes for the inversion-recovery multi-echo spin-echo, for a total of 46 minutes.

> *[Reviewer 2, Comment 3] In what format were the cells used for patch clamping, and were any controls done to ensure that characteristics of these cells were the same as those pelleted and imaged in the MRI studies? How long were the incubation times with ionic solutions in the patch clamp experiment? This information should likewise be added to the paper.*

[Reviewer 2, Response 3] We have clarified in the revised manuscript that SH-SY5Y cells were patch clamp-measured in their adherent state. On the other hand, the cells were dissociated from the culture plate and pelleted, so the experimental environments were not entirely identical. The patch clamp experiments involved a 20–30 minutes incubation period with the ionic solutions. We have included this information in the revised manuscript.

> *[Reviewer 2, Comment 4] Can the authors provide information about the mean cell size observed under each condition in their in vitro experiments?*

[Reviewer 2, Response 4] We did not directly quantify the mean cell size for each in vitro condition in this study, so we do not have corresponding data. However, we acknowledge that this information could provide valuable insights into potential mechanisms underlying the observed MR parameter changes. In future experiments, we plan to include direct cell-size measurements to further elucidate how changes in cell volume or hydration contribute to our MR findings.

> *[Reviewer 2, Comment 5] The ionic challenges used both in vitro and in vivo could also have affected cell permeability, with corresponding effects that would be detectable in diffusion weighted imaging. Did the authors examine this or obtain any results that could reflect on contributions of permeability properties to the contrast effects they report?*

[Reviewer 2, Response 5] We did not perform diffusion-weighted imaging and therefore do not have direct data regarding changes in cell permeability. We agree that incorporating diffusion-weighted measurements could help distinguish whether the MR parameters changes are driven primarily by membrane potential shifts, cell volume changes, or variations in permeability properties. We will consider these approaches in our future studies.

> *[Reviewer 2, Comment 6] Clearly, a faster stimulation method such as optogenetics, in combination with time-locked MRI readouts of the pelleted cells, would be more effective at demonstrating a useful relationship between cellular neurophysiology and MRI contrast in vitro. Can the authors present data from such an experiment? Is there any information they can present that documents the time course of observed responses in their experiments?*

[Reviewer 2, Response 6] In the current study, our methodology did not include time-resolved or dynamic measurements. While it may be possible to obtain indirect information about the temporal dynamics using T^2-weighted or MT-weighted imaging, such an experiment was beyond the scope of this work. However, we agree that an optogenetic approach with time-locked MRI acquisitions could help directly link cell physiology to MRI contrast, and we will explore this in future studies.

> *[Reviewer 2, Comment 7] The authors used a drug cocktail to suppress hemodynamic effects in the experiments of Figs. 5-6. What evidence is there that this cocktail successfully suppresses hemodynamic responses and that it also preserves physiological responses to the ionic challenges used in their experiments? Were analogous in vivo results also obtained in the absence of the cocktail?*

[Reviewer 2, Response 7] We appreciate the reviewer's concern regarding pharmacological suppression of hemodynamic effects. Although each component is known to inhibit nitric oxide synthesis, we did not directly measure the degree of hemodynamic suppression in this study. In addition, we cannot definitively confirm that these agents preserved the physiological responses to the ionic challenges. We have clarified these points in the revised manuscript and identified them as limitations of the study.

> *[Reviewer 2, Comment 8] Why weren't PSR results reported as part of the in vivo experimental results in Fig. 5? Does PSR continue to vary inversely to T2 in these experiments?*

[Reviewer 2, Response 8] In our current experimental setup, acquiring the T^2 map four times required 48 minutes, and extending the scan to include additional quantitative MT measurements for PSR would have significantly prolonged the scanning session. Given that these experiments were conducted on acutely craniotomized rats, maintaining stable physiological conditions for such a long period of time was challenging. Therefore, due to time constraints, we did not perform MT measurements and focused on T_2 mapping.

> *[Reviewer 2, Comment 9] The authors have established in vivo optogenetic stimulation paradigms in their laboratory and used them in the Toi et al. DIANA study. Were T2 or PSR changes observed in vivo using standard T2 measurement or T2-weighted imaging methods that do not rely on the DIANA pulse sequence they originally applied?*

[Reviewer 2, Response 9] Our current T_2 mapping experiments utilized a standard multi-echo spin-echo sequence, rather than the DIANA pulse sequence employed in our previous work. In this respect, the T_2 changes we observed in vivo do not rely on the specialized DIANA methodology.

> *[Reviewer 2, Comment 10] In the discussion section, the authors state that to their knowledge, theirs "is the first report that changes in membrane potential can be detected through MRI." This cannot be true, as their own Toi et al. Science paper previously claimed this, and a number of the studies cited on p.2 also claimed to detect close correlates of neuroelectric activity. This statement should be amended or revised.*

[Reviewer 2, Response 10] We appreciate the reviewer's comment. We have revised the discussion section of the manuscript to reflect the points raised by the reviewer.

> *[Reviewer 2, Comment 11] Because the current study does not actually demonstrate that changes in membrane potential can be detected by MRI, the authors should alter the title, abstract, and a number of relevant statements throughout the text to avoid implying that this has been shown. The title, for instance, could be changed to "Responses to depolarizing and hyperpolarizing ionic solutions measured by magnetic resonance imaging of excitable cells and rat brains," or something along these lines.*

[Reviewer 2, Response 11] We appreciate the reviewer's suggestions. We have revised the title, abstract, and relevant statements of the manuscript to clarify that our findings show MR-detectable responses to ionic solutions that are expected to modulate membrane potential, rather than demonstrating direct detection of membrane potential changes by MRI.

> *[Reviewer 2, Comment 12] The axes in Fig. 3 seem to be mislabeled. I think the horizontal axes are supposed to be membrane potential measured in mV.*

[Reviewer 2, Response 12] Thank the reviewer for finding an error. We have corrected the axis labels in Figure 3 to indicate membrane potential (in mV) on the horizontal axis.

> *[Reviewer 2, Comment 13] Since neither the experiments in Jurkat cells (Fig. 4) nor the in vivo MRI tests (Fig. 5-6) appear to have made in conjunction with membrane potential measurements, it seems like a stretch to refer to these experiments as involving manipulation of membrane potentials per se. Instead, the authors should refer to them as involving administration of stimuli expected to be depolarizing or hyperpolarizing. The "hyperpolarization" and "depolarization" labels of Fig. 4 similarly imply a result that has not actually been shown, and should ideally be changed.*

[Reviewer 2, Response 13] To prevent any misleading that membrane potential changes were directly measured in Jurkat cells or in vivo, we have revised the relevant text and figure labels.

> *[Reviewer 2, Comment 14] The changes in T2 and PSR documented with various K^+ challenges to Jurkat cells in Fig. 4 seem to follow a step-function-like profile that differs*

> *from the results reported in SH-SY5Y cells. Can the authors explain what might have caused this difference?*

[Reviewer 2, Response 14] We currently do not have a definitive explanation for why Jurkat cells exhibit a step-function-like response to varying K^+ levels, whereas SH-SY5Y cells show a linear response to log [K^+]. Experiments that include direct membrane potential measurements in Jurkat cells would help clarify whether this difference arises from genuinely different patterns of depolarization/hyperpolarization or from other factors. We have revised the revised manuscript to address this point.

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